



An Internship report

Of

Thiranex C++ ProgrammingVirtual Internship

**Submitted
To**

School of Computer Science and Engineering
IILM University, Greater Noida, U.P.

In partial fulfillment of the requirement of degree of
B.Tech CSE(Computer Science Engineering)

By

Roll Number of Student: 25SCS1003000929

Name of Student: HARSHIT PANDEY

Batch: 2026–2027

1. CANDIDATE'S DECLARATION

I, Harshit Pandey, do hereby declare that the internship report titled "**C++ Programming Internship Projects**" has been completed by me to fulfill the requirement for the award of the degree of Bachelor of Technology in CSE. All the references have been quoted and I have not taken as such any material from any other source. I affirm to you that this report is my own and has not been submitted to any other institute for any degree/diploma requirement and further I shall be solely responsible for any kind of copyright violation in this regard.

Student Name: Harshit Pandey

Roll No.: 25SCS1003000929

2. ACKNOWLEDGMENT

I am using this opportunity to express my gratitude to everyone who supported me throughout the Internship Program. I am thankful for their aspiring guidance, invaluable constructive criticism, and friendly advice during this work. I am sincerely grateful to them for sharing their truthful and illuminating views on a number of issues related to this work.

I express my gratitude to my mentors at **Thiranex (Skill Development & Future Tech)** for providing me with the opportunity to gain practical exposure through real-world software engineering projects.

I express my gratitude to my parents for their blessings.

Student Name: Harshit Pandey

Roll No.: 25SCS1003000929

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3. INTERNSHIP COMPLETION CERTIFICATE

(Placeholder for the Official Completion Certificate issued by Thiranex upon internship completion on 18 Sept 2026)

Offer letter of the Internship



Date: 19 August 2026

TO,
Harshit Pandey
Intern ID: THX-AUG1926-190

Subject: Offer Letter for Internship in C++ Programming

Dear **Harshit Pandey**,

Following our selection process, we are pleased to offer you an internship position at **Thiranex**. We believe your skills and enthusiasm will be a great addition to our team.

Internship Role	Intern - C++ Programming
Commencement Date	19 Aug 2026
Completion Date	18 Sept 2026
Work Mode	Remote / Project-Based

During this tenure, you will work on practical projects under industry mentorship. Your progress will be reviewed periodically, and your performance will determine the successful completion of the internship.

We look forward to a mutually beneficial learning journey. Please confirm your acceptance by starting your onboarding tasks.



This is an electronically generated document. No physical signature is required.
For verification, contact Thiranex Verification Cell.
www.thiranex.in



4. PROJECT DESCRIPTION

4.1 Introduction

During the internship period from **19 August 2026 to 18 September 2026**, I engaged as an **Intern - C++ Programming** at **Thiranex**. The internship emphasized hands-on implementation of Core C++ concepts, Object-Oriented Programming (OOP), File Handling, and Algorithmic Logic through four core project deliverables:

1. **Student Management System** (Due: 26 Aug 2026)
2. **Bank Management Application** (Due: 02 Sept 2026)
3. **Library Management System** (Due: 09 Sept 2026)
4. **Mini Game Project (Tic Tac Toe / Snake Game)** (Due: 16 Sept 2026)

4.2 Organization Profile

Thiranex (Skill Development & Future Tech) is an educational technology and skill development organization dedicated to equipping students with practical engineering capabilities. Operating on a remote, project-based work structure, Thiranex provides industry mentorship, structured task milestones, and evaluation pipelines to ensure interns acquire production-ready software development skills.

4.3 Problem Statement

Modern software engineering requires foundational expertise in memory safety, efficient data structures, file persistence, and modular software design. Standard academic courses often lack intensive hands-on application building. The key challenges addressed in this internship included:

- Persistent storage implementation without relational database engines using raw file streams (fstream).
- Object abstraction and secure data access using Object-Oriented encapsulation.
- Efficient console interface design utilizing menu-driven loops and dynamic state updates.

4.4 Project Objectives

The key objectives across the four assigned modules included:

- **Student Management System:** Build a persistent record system allowing CRUD (Create, Read, Update, Delete) operations using C++ file streams.
- **Bank Management Application:** Simulate core banking transactions (deposits, withdrawals, balance checks) enforcing encapsulation and secure record storage.
- **Library Management System:** Model book inventories and member lending transactions using C++ classes and search lookup algorithms.
- **Mini Game Project:** Build a terminal-based interactive game using 2D arrays, matrix manipulation, loop structures, and dynamic render cycles.

4.5 Scope of the Project

The project scope encompassed console-based applications targeted for native CLI environments. It focused heavily on data integrity, state management, algorithmic logic, and input validation without external heavy framework dependencies.

4.6 Technologies and Tools Used

- **Programming Language:** C++ (C++11 / C++17 Standards)
- **Development Environment (IDE):** VS Code / Code::Blocks / GCC Compiler
- **Version Control:** Git & GitHub
- **Libraries Used:** <iostream>, <fstream>, <string>, <vector>, <iomanip>, <cstdlib>

4.7 Methodology

The development lifecycle followed an iterative, milestone-driven approach:

1. **Requirement Analysis:** Understanding problem constraints and data model requirements.
2. **Object & Data Structure Design:** Defining structures (struct) and classes (class) with private members and public interfaces.
3. **Storage Strategy:** Implementing file operations (append, read-sequential, write-override) to ensure persistence.
4. **Logic Implementation:** Writing core algorithms for updates, search filtering, and state transitions.
5. **Testing & Debugging:** Verifying border conditions, dynamic state rendering, and input exception handling.

4.8 Expected Outcomes

All targeted deliverables were completed successfully and submitted on track prior to their respective due dates:

- **Student Management System:** Complete console app executing CRUD operations via persistent flat files.
- **Bank Management Application:** Secure system handling financial transactions and maintaining customer log persistence.
- **Library Management System:** Functional module supporting quick author/title searches and issue/return dynamic tracking.
- **Mini Game Project:** Dynamic game engine demonstrating real-time win/loss condition tracking and replay options.

4.9 Certificates of Completion & Communication Proofs

Offer Letter Details

- **Intern ID:** THX-AUG1926-190
- **Role:** Intern - C++ Programming
- **Tenure:** 19 Aug 2026 – 18 Sept 2026
- **Mode:** Remote / Project-Based



SKILL DEVELOPMENT & FUTURE TECH

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Harshit Pandey
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Milestone Submission Proofs

- **Task 1 (Student Management System):** Due 26 Aug 2026 — *Status: Submitted on Track / Completed*
- **Task 2 (Bank Management Application):** Due 02 Sept 2026 — *Status: Submitted on Track / Completed*
- **Task 3 (Library Management System):** Due 09 Sept 2026 — *Status: Submitted on Track / Completed*
- **Task 4 (Mini Game Project):** Due 16 Sept 2026 — *Status: Submitted on Track / Completed*

5. BIBLIOGRAPHY / REFERENCES

1. Geekforgeeks: OOPS with C++
2. Thiranex Skill Development & Future Tech. (2026). *Internship Program Task Briefs and Documentation*. <https://www.thiranex.in>